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A New Space-Time Model for Interacting Agents in the Financial Market

Master's Thesis in Financial Mathematics

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MASTER'S PROGRAMME IN FINANCIAL MATHEMATICS

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Abstract

In this thesis we present a new space-time model of interacting agents in the financial market. It is a combination of the Curie-Weiss model and a model introduced in Järpe [5]. We investigate properties such as the critical temperature and magnetization of the system. The distribution of the Hamiltonian function is obtained and a hypothesis test of independence is derived. The results are illustrated in an example based on real data.

Contents

Chapter 1

Introduction

1.1 The history of the Ising model

In 1920 professor Lenz of the University of Rostock in Germany suggested a model about paramagnetism and ferromagnetism and published it (see Niss [15]). Lenz had pointed at the magnetic turnover processes in solid paramagnetic materials in order to explain the Curie law, with an outlook to ferromagnetism. After Lenz moved from Rostock to Hamburg, he assigned the calculational tasks of his model to his student Ising, who wrote his results in his thesis of 1924 and in a paper of 1925. The main assumption that Ising made is that the nearby spins reach lower potential by pointing at the same direction (see Ellis [3]). The model became well-known as Ising, despite the fact that Ising wrote that the model really should be called Lenz-Ising model. The Ising model became one of the most studied models in modern physics. It was the major point to study phase transitions (see Domb [2]) and has been applied to a wide range of physical phenomena. Ising found a correct solution of the problem in one dimension and showed that the solution admits no phase transition in this case. Elevating the calculations to higher dimensional cases he made a few wrong conclusions which lead to no phase transitions in these cases (see Niss [15], Kobe [12]).

The existence of phase transition in the cases of two and three dimensions was stated in Peierls [16]. In 1941 Kramers and Wannier [13] found the exact location of the critical temperature for the Ising model on the square lattice for the two dimensional case. In 1944 Onsager [17] obtained a complete analytic solution for the two dimensional Ising model with zero external field.

1.2 Ising-type models and their applications

After the discovery of the critical temperature by P. Curie, Weiss developed a theory of ferromagnetism based on a spin system. It appears by replacement of the nearest-neighbor interaction of the Ising model by the assumption that at any site of the lattice each pair of spin variables interact with each other with exactly the same strength. Different variations of the Curie-Weiss model exist. Often models are considered to live on random graphs where each pair of sites are neighbors with probability p.

The Ising model became very popular not just because its being a tool explaining critical temperatures for which phase transitions occur in physical systems. It also has plenty of applications in the different sciences. Weidlich [18],[7] was the first who used Ising model to explain the polarization phenomena in sociology. It has also been adapted in economics to explain the diffusion of technical innovations. New technologies were considered to be the result of the interaction with neighboring firms. Into general equilibrium economics Ising model was introduced by Föllmer [6] in the following sense: there are several sources of influence on the price of the stock. One important component is real firm data; another one correlation between amount of buyers and sellers. The action of each trader (being a buyer or seller) was modeled as the value of the trader. Today information about trading on the markets is available e.g. in Internet for everyone. Therefore the assumptions in the model about influence of every trader are supported by the market system.

The Ising model is a simplification of a more general model, in which the energy function contains not just pairwise interaction between sites, but also interactions of bigger amount of the sites:

$$H = a_1 x_1 x_2 \cdot \ldots \cdot x_N + a_2 \sum_{i=1,\ldots,N} x_1 x_2 \cdot \ldots \cdot x_{i-1} x_{i+1} \cdot \ldots \cdot x_N + \ldots + a_N \sum_{i=1,\ldots,N} x_i,$$

where N is a number of sites. But despite that, the Ising model is strong and useful in various applications.

Kaizoji [9] introduced an Ising-type model of speculative activity as an attempt to explain bubbles and crashes in stock market. He introduced the market-maker, who adjusts the price on the market as a function of correlation of the buyers and sellers. Excess demand for stock at a period is $d\sum_{i=1}^{N} x_i$, where d is a size of the stock. Depending on the sign of this sum the market-maker raises or reduces the stock price. At equilibrium the amount of buyers is equal to the amount of sellers and $\sum_{i=1}^{N} x_i = 0$. As an example of using this model Kaizoji considered the Japan Crisis (1987–1992) and described it in terms of the critical temperature of the market.

Also other space-time models have been suggested. One interesting model based on the Ising model, was introduced in Järpe [5]. The Hamiltonian of this model contains the time dependence in a multiplicative form.
The model which is developed in this thesis has the same structure of the energy function as the energy function in Järpes model but absence of neighborhood structure like in the Curie-Weiss model, i.e. all traders can interact directly with each other. Also in this paper we introduce some results about critical temperature of the market in this model and analyze the distribution of the Hamiltonian.

In this section the background is given. In Section 2 the Ising model, the Curie-Weiss model and some other space and space-time models are considered. Also a method to find the critical temperature of the Curie-Weiss model is presented. In Section 3 the new space-time model is considered, the critical temperature for this model is derived and the distribution of the Hamiltonian is found. Also hypothesis tests for model parameters are derived, and an example based on real data is presented. The discussion of the results is in Section 4. Finally simulation program code may be found in the Appendix.

Chapter 2

Methods

2.1 The Ising model

The Ising model (also known as the Lenz-Ising model) is defined over a symmetric hypercube Λ of the D-dimensional integer lattice ($D \in \{1, 2, \ldots\}$). A particle is assigned to each lattice site. The sites are connected by bonds. Two connecting sites are called neighbors. The set of all neighbors of the site i is called *neighborhood* of i. If the sites i and j are neighbors then we write $i \sim j$, where the neighborhood relation is subject to the conditions: $i \nsim i$ and $i \sim j \Leftrightarrow j \sim i$. Each site of the lattice can adopt two states, $X_i \in \{-1, 1\}, i = 1, 2, \ldots, N$, where N is a number of the sites in the lattice. Sometimes these states are referred to as spins and the values are referred to as down and up respectively. A configuration of the model is a specification of the spin for each lattice site. With each configuration $x = \{x_i : i = 1, 2, \ldots, N\}$ a Hamiltonian or interaction energy is connected

$$H(x) \;=\; -\alpha \sum_{i,j \in \Lambda : i \sim j} x_i x_j - h \sum_{i \in \Lambda} x_i. \tag{2.1}$$

The first sum extends over all distinct neighbor pairs of sites in Λ. The number α is the strength of the nearest neighbor interaction; the same for the all pairs. The second sum is over all lattice sites and h is a real number which is the strength of the externally applied magnetic field. Thus, if the spins of two nearest neighbors in the configuration have the same orientation (both up or both down), then this pair contributes an energy $-\alpha$ to the total energy. If they have opposite orientations, they contribute an energy α. Thus, if α is positive then neighbors have the tendency to be in the same state (so-called *attractive case*). If α is negative then neighbors have the tendency to be in the opposite state (*repulsive case*). So for the attractive case the first term of Equation (2.1) contributes minimum energy when all the sites are in the same state.

Suppose that the energy of each configuration x has been determined. The probability, p, that the system has configuration x with energy H, according to a fundamental assumption of equilibrium statistical mechanics, is given by the so-called *Boltzman factor*

$$p(x) = \frac{1}{Z} e^{\beta H(x)}, \tag{2.2}$$

where $Z = \sum_x e^{\beta H(x)}$ is a partition function since the sum of the probabilities for all states has to equal 1. The parameter β (depends on α) is the *temperature* of the market, shows the randomness of the trading.

2.2 Sufficient statistic

Proposition 1. *The Hamiltonian H is a sufficient statistic for the temperature β.*

Proof. This is clear from the factorization theorem because the distribution is a member of the exponential family. □

2.3 The Curie-Weiss model

The Curie-Weiss model is a simple modification of the Ising model. It allows all agents in the system to interact with each other with a constant strength. The Hamiltonian in the Curie-Weiss model has the form

$$H(x) \;=\; \alpha \sum_{i,j \in \Lambda : i<j} x_i x_j - h \sum_{i \in \Lambda} x_i. \tag{2.3}$$

The first sum now extends over all pairs of the sites in Λ.

2.4 Interpretation for financial applications

Let us consider the Curie-Weiss model now. Assume that a financial market consists of N traders $i = 1, 2, \ldots, N$. Every trader can buy fixed amount, d, of stock or sell the same amount. In the first case *"the decision of trader i"* is $X_i = 1$, in the second case $X_i = -1$. Then $X = (X_1, X_2, \ldots, X_N)$ represents the investment attitude of the market.

The Law of demand and supply says that if amount of buyers is larger than amount of sellers then there is excess demand on the market and consequently the price goes up. If the amount of sellers is larger, then there is excess supply and the price goes down. So to get profit each trader should try to predict the behavior of other traders and choose the same strategy as the majority before trading.

Another important factor of investment's attractiveness is *investment environment*

$$s = \frac{\{\text{profit}\}}{\{\text{total capital}\}} - \{\text{interest rate on alternative asset (bond)}\}, \tag{2.4}$$

If s increases then it is time to buy, otherwise it is time to sell. This may be expressed for trader i in terms of minimizing a disagreement function

$$q_i(x) \;=\; -\;\frac{1}{2}\sum_{j=1}^{N} a_{ij}x_ix_j - b_isx_i, \tag{2.5}$$

where a_{ij} is strength of influence of the trader j on i; b_i is strength of the reaction of i upon the change of s. The coefficient a_{ij} can be also interpreted as follows: if there is no crisis or bubble in the market then $|a_{ij}|$ is small, and the actions of the traders are weakly dependent, otherwise $|a_{ij}| > 0$ and the traders i and j are more strongly affected by each other.

The disagreement function of all market is

$$H(x) = \sum_{i=1}^{N} q_i(x) = -\frac{1}{2}\sum_{i=1}^{N}\sum_{j=1}^{N} a_{ij}x_ix_j - \sum_{i=1}^{N} b_isx_i, \tag{2.6}$$

which has to be minimized.

2.5 Motivation for assuming the Boltzman distribution

Assume that the variable X characterizes the state of the system. We may enumerate the distinct configurations of our system

$$x^k = (x_1^k, x_2^k, \dots, x_N^k),\ k = 1, \dots, K,$$

where $x_i^k \in \{-1, 1\}$. In this case

$$\begin{aligned} &x^1 = (-1, -1, \dots, -1),\\ &x^2 = (-1, -1, \dots, 1),\\ &\vdots\\ &x^K = (1, 1, \dots, 1). \end{aligned}$$

If we count the number of states we get $K = \#\{-1,1\}^N = 2^N$.

Let $p(x^k) = P(X = x^k)$ be the probability of observing the state x^k. Obviously we have that $0 \leq p(x^k) \leq 1$ and $\sum_{k=1}^{K} p(x^k) = 1$. Further we assume that all states x^k are possible (i.e. have positive probability) $0 < p(x^k) < 1$.

We wish to minimize the entropy of the system, which is a measure of uncertainty

$$Q = -\sum_{i=1}^{K} p(x^k)\ln(p(x^k)).$$

So, if all configurations are equally likely $(p(x^1) = \ldots = p(x^K))$ then the uncertainty is large and the entropy Q attains its maximum

$$\frac{\partial Q}{\partial x^k} = \sum_{i=1}^{K} p'(x^k)(\ln(p'(x^k)) + 1) = 0.$$

If we minimize the entropy the level of uncertainty is the smallest.

Thus the optimization problem (2.6) is formalized by

$$E_P(H(X)) = \sum_{k=1}^{K} p(x^k)H(x^k) \tag{2.7}$$

with respect to the measure P.

The solution of this problem is the well known Boltzmann distribution

$$p(x^k) = \frac{1}{Z}e^{-\beta H(x^k)},\ k = 1,\ldots,K, \tag{2.8}$$

where $Z = \sum_{k=1}^{K} e^{-\beta H(x^k)}$ and β is the market temperature, that describes the degree of randomness in the behavior of the traders.

2.6 Mean field theory

For calculation of the solution we can use the mean field approximation. Let us introduce *magnetization* of the system $M = \frac{1}{N}\sum_{i=1}^{N} X_i$, and assume that $a_{ij} = a,\ b_i = b\ \forall i,j = 1\ldots N$, then the disagreement function of the state x is

$$H(X) \approx -\frac{1}{2}\sum_{i=1}^{N} aNMx_i - \sum_{i=1}^{N} bsX_i.$$

Using the fact that mean value of $H(x)$ is

$$E(H(X)) = -\frac{\partial \ln Z}{\partial \beta},$$

the solution is

$$M \;=\; \tanh(\beta M \;+\; \alpha s), \tag{2.9}$$

where $\beta = \mu a N$ is called the *bandwagon coefficient.* It shows that traders tend to have the same attitude as prediction of the average. The paramerer $\alpha = \mu b$ is known as the *coefficient of the change of the investment environment.*

2.7 Stationarity

For the subsequent inferential goals, conditions for existance of a stationary distribution is of the essence. There are conditions on the parameters of the model and they are of vital importance for general statements about the model. We should find intervals for the parameters α and β, where $M = \tanh(\beta M + \alpha s_t)$. Analysis shows that if $0 < \beta < 1\ \forall \alpha s$ then $M = 0$ is an unique solution. If β is bigger than 1, then so-called *"Bear"* and *"Bull"* markets appear. This transition is called a second-order phase transition. If $0 < \beta < 1$ and $\alpha \neq 0$, then prices go up and down slowly. If $\beta > 1$, $\alpha \neq 0$ and $|s| < s^*$ there is an increased risk that a speculative bubble and market crash occurs. If $\beta > 1$, $\beta \neq 0$ and $|s| = s^*$ then first-order phase transition appears. If $|s|$ is bigger than s^* the bull or bear market appears.

2.8 The critical temperature

Let us consider a market of N traders. Every member, i, of the market has to make a "Trader's decision", of whether to buy ($X_i = 1$) or to sell ($X_i = -1$) the shares. Every trader knows the opinions of other traders, so the decision is under influence of the others. The system of N traders has 2^N states. The set of all states of the system is denoted by $\aleph = \{-1, 1\}^N$. We assume a zero external field in this model.

In Hartmann [8] the equation

$$M \;=\; \tanh(M\beta) \tag{2.10}$$

is considered, where $M = \frac{1}{N}\sum_i X_i$ represents *magnetization.* This equation shows the behavior of the system and it is the key to find the critical temperarure of the market. Let us consider the method how Equation 2.10 can be obtained.

The magnetization takes values in

$$\mathcal{M} \;=\; \left\{-1, -1 + \tfrac{2}{N}, -1 + \tfrac{4}{N}, \ldots, +1\right\}.$$

Let us fix configuration x of the system with magnetization m and Hamiltonian $H(x)$. The Hamiltonian function is

$$\begin{aligned} H(x) &= -\frac{1}{N}\sum_{i=1}^{N-1}\sum_{j=i+1}^{N} x_i x_j \\ &= -\frac{1}{2N}\left(\sum_{i=1}^{N} x_i\right)^2 + \frac{1}{2N}\sum_{i=1}^{N} x_i^2 \\ &= -\tfrac{N}{2}m^2 + \mathcal{O}(1). \end{aligned}$$

The partition function is

$$Z = \sum_{x\in\aleph} e^{-\beta H(x)} = \sum_{x\in\aleph}\sum_{m\in\mathcal{M}} \delta_{mx} e^{\beta N m^2/2}.$$

Here let us change the order of the sums so that

$$Z = \sum_{m\in\mathcal{M}}\left(\sum_{x\in\aleph}\delta_{mx}\right) e^{\beta N m^2/2}.$$

Now we will calculate the number of configurations leading to the fixed magnetization m. If we denote by $n_{+1} = \frac{N}{2} + \frac{\sum x_i}{2}$ the number of the traders who want to buy (i.e. they are in state $'+1'$) then this implies that $n_{-1} = N - n_{+1} = \frac{N}{2} - \frac{\sum x_i}{2}$ traders want to sell (i.e they are in state $'-1'$). This will correspond to the magnetization

$$m = \frac{\frac{N}{2} + \frac{\sum_{i=1}^{N} x_i}{2} - \left(\frac{N}{2} - \frac{\sum_{i=1}^{N} x_i}{2}\right)}{N}.$$

The total number of all configurations leading to the magnetization m is

$$\mathcal{N}(m) = \sum_{x\in\aleph}\delta_{mx} = \binom{N}{n_{+1}} = \binom{N}{N\frac{1+m}{2}} = \frac{N!}{(N\frac{1+m}{2})!(N\frac{1-m}{2})!}.$$

Then, by using Stirling's formula for large K: $\ln(K!) \approx K\ln(K) - K$, we obtain

$$\begin{aligned} \binom{N}{aN} &= \frac{N!}{(aN)!((1-a)N)!} \\ &\approx e^{N\ln N - N - (aN)\ln(aN) + aN - ((1-a)N)\ln((1-a)N) + (1-a)N} \\ &= e^{-Na\ln(a) - N(1-a)\ln(1-a)}. \end{aligned}$$

So, we have

$$\mathcal{N}(m) \;\approx\; e^{-N\frac{1+m}{2}\ln\frac{1+m}{2}-N\frac{1-m}{2}\ln\frac{1-m}{2}}. \tag{2.11}$$

The partition function can be replaced by integration for $N \gg 1$

$$\begin{aligned} Z &\approx \int_{-1}^{1} e^{-N\left(\frac{1+m}{2}\ln\frac{1+m}{2}+\frac{1-m}{2}\ln\frac{1-m}{2}-\frac{\beta}{2}m^2\right)}dm \\ &= \int_{-1}^{1} e^{-Nf(m)}dm, \end{aligned}$$

where

$$f(m) = \frac{1+m}{2}\ln\frac{1+m}{2}+\frac{1-m}{2}\ln\frac{1-m}{2}-\frac{\beta}{2}m^2.$$

To calculate this integral we will use Saddle-point approximation (see Erdelyi [4]). Assume that m_0 is the unique global minimum of the function $f(m)$. Taylor expansion around m_0 will have form

$$f(m) = f(m_0) + f'(m_0)(m-m_0) + \frac{1}{2}f''(m_0)(m-m_0)^2 + \mathcal{O}((m-m_0)^3).$$

Since $f(m)$ has a global minimum at m_0, function $-f(m)$ has a global maximum at m_0. Because of the large value of N this maximum will just increase and become much bigger than other values of the function. The value of $e^{-Nf(m)}$ is always positive and the value of the integral we can aproximately find through the value of this global minimum.

At m_0 the first derivative of f vanishes. So, the function $f(m)$ can be approximated as

$$f(m) \approx f(m_0) + \frac{1}{2}f''(m_0)(m-m_0)^2,$$

for m close to m_0. Now we use these assumptions for approximation of the partition function Z

$$\begin{aligned} Z &= \int_{-1}^{1} e^{(-Nf(m))}dm \\ &\approx e^{-Nf(m_0)}\int_{-1}^{1} e^{-\frac{1}{2}Nf''(m_0)(m-m_0)^2}dm. \end{aligned}$$

We remember that

$$\int_{-\infty}^{\infty} e^{-\frac{x^2}{2a}}dx = \sqrt{2\pi a},$$

so if $N \to \infty$, then

$$Z \approx e^{-Nf(m_0)} \cdot \sqrt{\frac{2\pi}{Nf''(m_0)}}.$$

Now we are searching for a magnetization m_0 which is the zero of the first derivative of function $f(m)$

$$0 = f'(m_0) = \frac{1}{2}\ln\left(\frac{1+m_0}{2}\right) + \frac{1}{2} - \frac{1}{2}\ln\left(\frac{1-m_0}{2}\right) - \frac{1}{2} - m_0\beta,$$

or

$$\frac{1}{2}\ln\frac{1+m_0}{1-m_0} = m_0\beta,$$

$$e^{m_0\beta} = \sqrt{\tfrac{1+m_0}{1-m_0}}, \quad e^{-m_0\beta} = \frac{1}{\sqrt{\tfrac{1+m_0}{1-m_0}}}.$$

Then since $\tanh(m) = \frac{e^m - e^{-m}}{e^m + e^{-m}}$, we have that

$$\tanh(m_0\beta) = \frac{\sqrt{\tfrac{1+m_0}{1-m_0}} - \frac{1}{\sqrt{\tfrac{1+m_0}{1-m_0}}}}{\sqrt{\tfrac{1+m_0}{1-m_0}} + \frac{1}{\sqrt{\tfrac{1+m_0}{1-m_0}}}} = \frac{\frac{1+m_0}{1-m_0} - 1}{\frac{1+m_0}{1-m_0} + 1} = \frac{\frac{1+m_0-1+m_0}{1-m_0}}{\frac{1+m_0+1-m_0}{1-m_0}} = \frac{2m_0}{2} = m_0.$$

Since this equality holds for any configuration we have Equation (2.10). ■

We now have to analyze Equation (2.10) and find the values of β which correspond to the stationarity of the system. The easiest way to do it is to consider the graphics of the function $g_1(m) = \tanh(\beta m)$ and $g_2(m) = m$, where $m \in \mathcal{M}$. We can see in Figures 2.1 and 2.2 that the functions $g_1(M)$ and $g_2(M)$ have one or 3 points of intersection depending on the value of β. If $\beta = 1$, then Equation (2.10) has two roots.

Let us now determine which of these roots that is global minimum of the function f.

When $\beta < 1$ the only root $m = 0$ is also a global minimum of the function f (Figure 2.3). If $\beta > 1$ two symmetrical roots correspond to global minima. They are a *bull* and *bear* markets parameter interval boundaries. The root $m = 0$ is a global maximum of the function f. Thus there is magnetism on the market in the case when $\beta > 1$ (Figure 2.4).

So, for the Curie-Weiss model the critical value of β is 1.

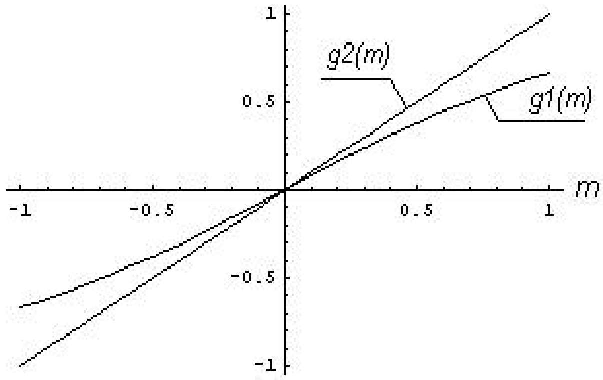

Figure 2.1: *When* β *is smaller than* 1 *Equation (2.10) has one solution.*

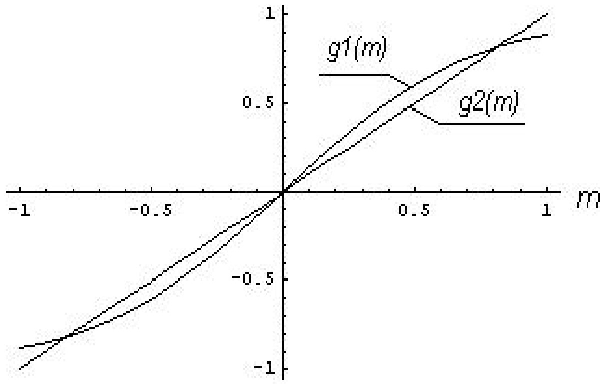

Figure 2.2: *When* β *is bigger than* 1 *Equation (2.10) has three solutions.*

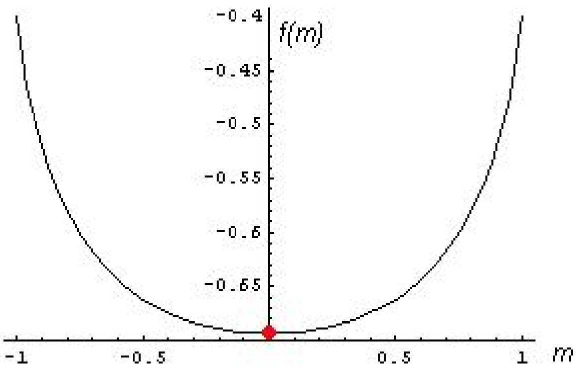

Figure 2.3: *When* β *is smaller than* 1 *Equation (2.10) has one solution.*

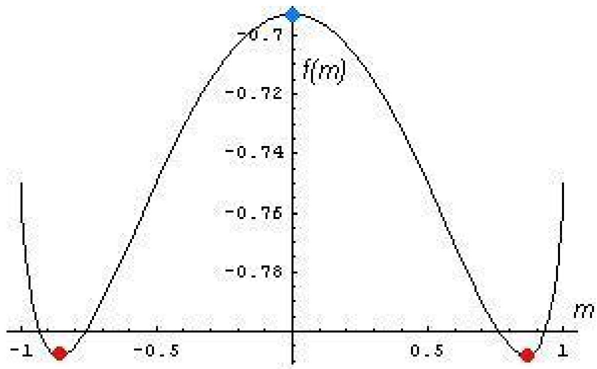

Figure 2.4: *When β is bigger than 1 Equation (2.10) has three solutions.*

2.9 A new space-time model

In Järpe [5] a model which possesses both spatial and time dependence is introduced. The state of a site in a lattice is depending on the states of its nearest neighbors and on the global degree of clustering of the previous pattern. The transition probability of the configuration of the system is

$$p_x(x|x') = p(X_t = x|X_{t-1} = x') = Z_{x'}^{-1} \exp(\beta R(x, x')),$$

where the Hamiltonian is a function of the previous configuration, x', and the present, x, in the form $R(x, x') = H(x)H(x')$, where $H(x) = -\frac{1}{N}\sum_{i\sim j} x_i x_j$ is a Hamiltonian, x' is configuration of the system at the previous moment of the time and $Z_{x'} = \sum_{y=1}^{N^2} e^{\beta H(y)H(x')}$ is the normalising constant.

Now let us introduce a model for volatility clustering with the similar structure as in the model above but with the sum in the Hamiltonian over all pairs like in the Curie-Weiss model: $H(x) = -\frac{1}{N}\sum_{i=1}^{N-1}\sum_{j=i+1}^{N} x_i x_j$. Each trader in this model is directly dependent of all the other traders and also on the magnetization at the previous moment of the time. The partition function is

$$Z = \sum_{x_t \in \aleph} e^{\beta H(x_t)H(x_{t-1})/N} = \sum_{x_t \in \aleph}\sum_{m \in \mathcal{M}} \delta_{m x_t} e^{\beta N m^2(t) m^2(t-1)/4}.$$

So, in this model in the transition probabilities we have in exponent also division by N

$$p_x(x|x') = p(X_t = x|X_{t-1} = x') = Z_{x'}^{-1} \exp(\beta H(x)H(x')/N).$$

We need it to use the Saddle-point approximation later on.

Chapter 3

Results

3.1 Theoretical Results

3.1.1 The critical temperature of the new space-time model

We have a new model with the partition function

$$Z = \sum_{x_t \in \aleph} e^{\beta H(x_t) H(x_{t-1})/N} = \sum_{x_t \in \aleph} \sum_{m \in \mathcal{M}} \delta_{m x_t} e^{\beta N m^2(t) m^2(t-1)/4}.$$

Proposition 2. *$H(X_t)$ conditional on the previous state, $H(X_{t-1})$, is sufficient statistic for β.*

Proof. This is clear from the factorization theorem because the transition probability $p_x(x|x')$ is a member of the exponential family. □

We want now to obtain the critical temperature of this model.

Theorem 1. *The critical temperature of the new model is $\beta = 2$.*

Proof. As in the calculations for the Curie-Weiss model we can change summing over $\mathcal{M}$ and $\aleph$,

$$Z = \sum_{m \in \mathcal{M}} \left(\sum_{x_t \in \aleph} \delta_{m x_t} \right) e^{\beta N m^2(t) m^2(t-1)/4}.$$

The value $\sum_{x_t \in \aleph} \delta_{m x_t}$ will be the same as before (see Equation (2.11))

$$\sum_{x_t \in \aleph} \delta_{m x_t} = e^{-N \frac{1+m(t)}{2} \ln \frac{1+m(t)}{2} - N \frac{1-m(t)}{2} \ln \frac{1-m(t)}{2}}.$$

Therefore we have

$$Z = \int_{-1}^{1} e^{-N\left(\frac{1+m(t)}{2}\ln\frac{1+m(t)}{2}+\frac{1-m(t)}{2}\ln\frac{1-m(t)}{2}\right)+N\frac{\beta}{4}m^2(t)m^2(t-1)}dm(t).$$

This integral has the form

$$\int_a^b \exp\left(-Nf(m)\right)dm, \tag{3.1}$$

where N is large. Thus we can use the Saddle-point approximation to calculate Z.

As in the static (not dynamic) model we have to find the global minimum of the function f, where f now is

$$f(m(t),m(t-1)) = \tfrac{1+m(t)}{2}\ln\tfrac{1+m(t)}{2}+\tfrac{1-m(t)}{2}\ln\tfrac{1-m(t)}{2}-\tfrac{\beta}{4}m^2(t)m^2(t-1).$$

But now the function f depends on two variables, $m(t)$ and $m(t-1)$. Partial differentiation f with respect to $m(t)$ yields

$$f'_{m(t)}(m(t),m(t-1)) = \tfrac{1}{2}\ln\left(\tfrac{1+m_0(t)}{2}\right)+\tfrac{1}{2}-\tfrac{1}{2}\ln(\tfrac{1-m_0(t)}{2})-\tfrac{1}{2}-m_0(t)m_0^2(t-1)\beta = 0.$$

Then we have

$$\frac{1}{2}m(t)m^2(t-1)\beta = \ln\left(\sqrt{\frac{1+m(t)}{1-m(t)}}\right),$$

or

$$m(t) \;=\; \tanh\left(\frac{m(t)m^2(t-1)\beta}{2}\right). \tag{3.2}$$

Equation 3.2 holds for any configurations x_t and x_{t-1} with magnetization M_t and M_{t-1} correspondently. Thus we have

$$M(X_t) \;=\; \tanh\left(\frac{M(X_t)M^2(X_{t-1})\beta}{2}\right). \tag{3.3}$$

Partial differentiation of f with respect to $m(t-1)$ yields

$$f'_{m(t-1)}(m(t),m(t-1)) \;=\; -\frac{\beta}{2}m^2(t)m(t-1) \;=\; 0. \tag{3.4}$$

Now we have to analyse the function f with help of Equation (3.2) and Equation (3.4) in dimention 3. In Figure 3.1 and Figure 3.2 we can see graphics of the function f in the cases when β is equal 1 and 5 correspondently. In Figure 3.1 the value $m(t) = 0$ is a global minimum of f $\forall m(t-1)$. In

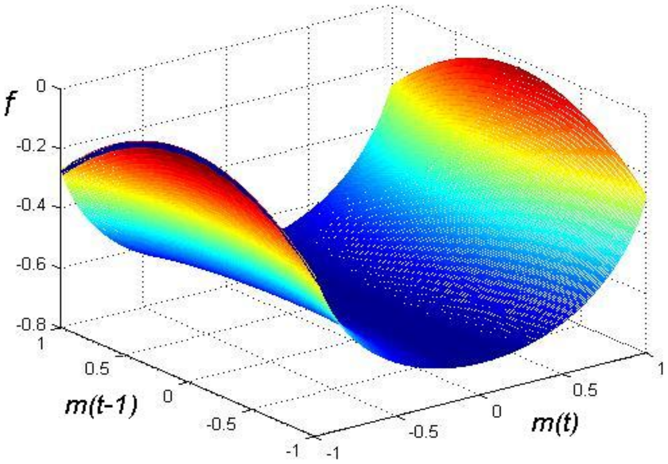

Figure 3.1: *When β is smaller than 2 function f has the global minimum at $m(t) = 0$.*

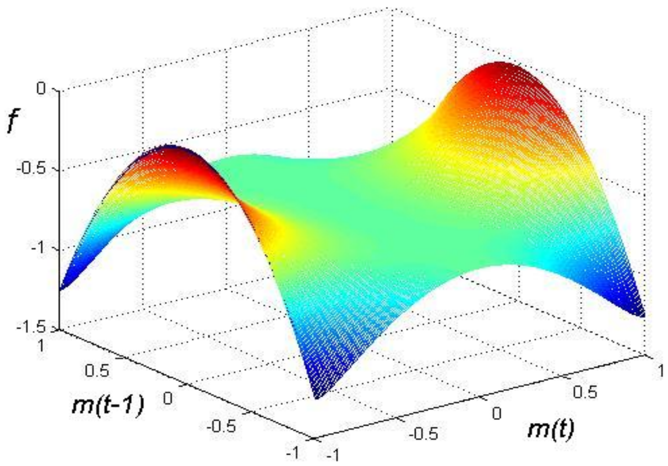

Figure 3.2: *When β is bigger than 2 function f has no global minimum at $m(t) = 0$.*

Figure 3.2 the value $m(t) = 0$ is not the global minimum. Depending on the value β all graphics of the function f have the form of Figure 3.1 either Figure 3.2.

Let us consider Figure 3.2 more carefully. If we fix some value $m_1(t-1)$ then function $f(m_1(t-1), m(t))$ will have or not have a global minimum at the point $m(t) = 0$ depending on which value of $m_1(t-1)$ was fixed.

If $|m(t-1)| < m_c(t-1)$ then $m(t) = 0$ is global minimum, otherwise not. Let us calculate this value $m_c(t-1)$. We need to find the interval where the second partial derivatives of f over $m(t)$ will positive at the point $m(t) = 0$. We get

$$f'_{m(t)}(m(t), m(t-1)) = \tfrac{1}{2}\ln\left(\tfrac{1+m_0(t)}{2}\right) + \tfrac{1}{2} - \tfrac{1}{2}\ln(\tfrac{1-m_0(t)}{2}) - \tfrac{1}{2} - m_0(t)m_0^2(t-1)\beta,$$

and

$$\begin{aligned} f''(m(t-1), m(t))\,\big|_{m(t)=0} &= \tfrac{1}{2}\tfrac{1-m(t)}{1+m(t)}\left(\tfrac{1-m(t)+1+m(t)}{(1-m(t))^2}\right) - \tfrac{\beta}{2}m^2(t-1)\,\big|_{m(t)=0} \\ &= \frac{1}{1-m^2(t)} - \frac{\beta}{2}m^2(t-1)\,\big|_{m(t)=0} \\ &= 1 - \frac{\beta}{2}m^2(t-1) > 0. \end{aligned}$$

Or

$$|m(t-1)| < \sqrt{\frac{2}{\beta}}. \tag{3.5}$$

Thus we have condition for the stationarity

$$|M(X_{t-1})| < \sqrt{\frac{2}{\beta}}. \tag{3.6}$$

If for example $\beta = 2$, condition (3.5) become $|m(t-1)| < 1$. Thus $\beta = 2$ is the critical value.

We also have the second oportunity to obtain this critical value. We can compare Equation (3.3)

$$M(X_t) = \tanh\left(\frac{M(X_t)M^2(X_{t-1})\beta}{2}\right),$$

with the Equation (2.10) of the behavior of the classical Curie-Weiss model

$$M = \tanh(M\beta).$$

In the moment t the value $M(t-1)$ is fixed. Thus we can rename expression $\frac{M^2(X_{t-1})\beta}{2}$ as a new constant β_1. Then we will receive Equation (2.10) and could use the condition of stationarity for the Curie-Weiss model: the region of stationarity is $\beta_1 < 1$. So we have condition

$$\frac{M^2(X_{t-1})\beta}{2} < 1,$$

or

$$|M(X_{t-1})| < \sqrt{\frac{2}{\beta}}.$$

We have obtained the same result.

$\square$

Conclusion: The unconditional region of stationarity is $\beta < 2$.
If $\beta > 2$ then we have stationarity in the case when $|M(X_{t-1})| < \sqrt{\frac{2}{\beta}}$. Thus the critical value β_c is equal to 2.

When $N \to \infty$ the partition function has the form

$$Z \approx \exp(-Nf(m_0(t), m(t-1))) \cdot \sqrt{\frac{2\pi}{Nf''(m_0(t), m(t-1))}}.$$

We know the function f and we received the form of the f''

$$f(m(t), m(t-1)) = \tfrac{1+m(t)}{2} \ln \tfrac{1+m(t)}{2} + \tfrac{1-m(t)}{2} \ln \tfrac{1-m(t)}{2} - \tfrac{\beta}{4} m^2(t) m^2(t-1),$$

$$f(m(t), m(t-1)) \big|_{m(t)=0} = \ln \frac{1}{2},$$

$$f''(m(t-1), m(t)) \big|_{m(t)=0} = 1 - \frac{\beta}{2} m^2(t-1).$$

Then

$$Z \approx \exp(-N \ln \frac{1}{2}) \cdot \sqrt{\frac{2\pi}{N\left(1 - \frac{\beta}{2} m^2(t-1)\right)}}.$$

3.1.2 Hamiltonian distribution with independent traders

Now we consider the function H. We are interested in analysis of dependence between X_i and X_j for $i, j = 1, \ldots, N$.

Theorem 2. *Let the variables $X_i, i \in \{1, \ldots, N\}$ independently of each other take its values in $\{-1, 1\}$ with equal probability $\frac{1}{2}$. Then all non-identical pairwise products are independent, i.e. $X_i X_j \perp X_l X_k$ if $i \neq j, k, l$ or $j \neq i, k, l$ for any dimension N.*

Proof. To prove that we will use method of induction with respect to the number of traders:

1. Consider the case $N = 3$. Without losing generality we will prove that XY is independent of XZ if X, Y and Z are independence. We need to prove that $P(XY = a; XZ = b) = P(XY = a)P(XZ = b)$, where a and b could be 1 or -1.

   (a) Let us calculate both parts for $a = 1$ and $b = 1$

$$P(XY = 1; XZ = 1) =$$

$$\begin{aligned}
&= P(XY = 1; XZ = 1\,|X = 1)P(X = 1)+\\
&\quad +P(XY = 1; XZ = 1\,|X = -1)P(X = -1)\\
&= P(Y = 1; Z = 1)P(X = 1) + P(Y = -1; Z = -1)P(X = -1)\\
&= P(Y = 1)P(Z = 1)P(X = 1) +\\
&\quad +P(Y = -1)P(Z = -1)P(X = -1)\\
&= p^3 + (1 - p)^3\\
&= p^3 + 1 - 3p + 3p^2 - p^3\\
&= 3p^2 - 3p + 1,
\end{aligned}$$

$$P(XY = 1)P(XZ = 1) =$$

$$\begin{aligned}
&= (P(Y = 1\,|X = 1)P(X = 1) + P(Y = -1\,|X = -1)\,P(X = -1)) \cdot\\
&\quad \cdot(P(Z = 1\,|X = 1)\,P(X = 1) + P(Z = -1|X = -1)P(X = -1))\\
&= (P(Y = 1)P(X = 1) + P(Y = -1)P(X = -1)) \cdot\\
&\quad \cdot(P(Z = 1)P(X = 1) + P(Z = -1)P(X = -1))\\
&= (p^2 + (1 - p)^2)(p^2 + (1 - p)^2)\\
&= (p^2 + (1 - p)^2)^2\\
&= (2p^2 - 2p + 1)^2.
\end{aligned}$$

   Under assumption that $p = \frac{1}{2}$ we have that for the case $a = 1\,, b = 1$ that variables XY, XZ are independent

$$\begin{aligned}
P(XY = 1; XZ = 1) &= 3p^2 - 3p + 1\\
&= \frac{1}{2}\\
&= (2p^2 - 2p + 1)^2\\
&= P(XY = 1)P(XZ = 1).
\end{aligned}$$

(b) To prove the others combinations of a and b we can use the properties of the probabilities. Let us call A and B the probability $P(XY = 1)$ and $P(XZ = 1)$ correspondently. Then $A^C = P(XY = -1)$, $B^C = P(XZ = -1)$. We obtained that $A \perp B$. Thus $A^C \perp B$, $A \perp B^C$ and $A^C \perp B^C$.

So we have for $N = 3$ independence of all district pair products of Bernoulli variables, i.e. $X_i X_j \perp X_l X_k$ if $i \neq j, k, l$ or $j \neq i, k, l$.

2. Assume that $X_i X_j X_l X_k$ for all $i, j, k, l \in \{1, \ldots, L\}$, such that $i \neq j, k, l$ or $j \neq i, k, l$.

3. Let us prove independence for the case $N = L+1$. If numbers i, j, k and l are in $\{1, \ldots, L\}$, then independence of the pair products is obvious because of the assumption in previous case. So we need to consider the case, when one number, let us say i, is $L + 1$. There are two cases exists for analysis. We should consider pairs $X_i X_{L+1}$, $X_i X_j$ and $X_i X_{L+1}$, $X_j X_{L+1}$. In the first case we can change X_i to X, X_{L+1} to Y and X_j to Z then there will be the same calculations as in the case $N = 3$. In the second case independence is proved by replacement X_{L+1} by X, X_i by Y and X_j by Z.

So for any N we have that in the case when $P(X_i = -1) = P(X_i = 1) = \frac{1}{2}$, the pairwise products $X_i X_j$ and $X_l X_k$ are independent whenever $i \neq j, k, l$ or $j \neq i, k, l$.

□

In this paper, we consider a model where a and b are decisions of the traders. For every trader the probability of their decision to 'buy' or 'sell' is $\frac{1}{2}$ unconditional of the other traders.

Now let us consider the set of values of the mean field $M = \frac{1}{N} \sum_{i=1}^{N} X_i$. For this we have the sample space or state space

$$M \in \left\{ -1, -1 + \frac{2}{N}, -1 + \frac{4}{N}, \ldots, 1 \right\}.$$

meaning that M can be in $N + 1$ states. Consider now values of M^2. If N is even, then one value of M is 0 and

$$M^2 \in \left\{ 1, \left(1 - \frac{2}{N}\right)^2, \left(1 - \frac{4}{N}\right)^2, \ldots, 0. \right\},$$

which implies that M^2 can be in $\frac{N}{2}+1$ different states. If N is odd then 0 is not the value of M and

$$M^2 \in \left\{1, \left(1-\frac{2}{N}\right)^2, \left(1-\frac{4}{N}\right)^2, \ldots, \frac{1}{N^2}\right\},$$

and M^2 can be in $\frac{N+1}{2}$ different states.

We remember that $H(X) = -\frac{1}{N}\sum_{i<j} X_i X_j = -\frac{N}{2}M^2 + \frac{1}{2}$. That is why H has the following values. If N is even then

$$H \in \left\{\frac{1-N}{2}, \frac{1}{2} - \frac{N}{2}\left(1-\frac{2}{N}\right)^2, \frac{1}{2} - \frac{N}{2}\left(1-\frac{4}{N}\right)^2, \ldots, \frac{1}{2}\right\}.$$

If N is odd then

$$H \in \left\{\frac{1-N}{2}, \frac{1}{2} - \frac{N}{2}\left(1-\frac{2}{N}\right)^2, \frac{1}{2} - \frac{N}{2}\left(1-\frac{4}{N}\right)^2, \ldots, \frac{1}{2} - \frac{1}{2N}\right\}.$$

Lemma 1. *If $P(X_i = 1) = p$ and $\{X_i\}$, $i = 1, \ldots, N$ are independent, then $\frac{NM+N}{2} \in Bin(N,p)$, where $M = \frac{1}{N}\sum_{i=1}^{N} X_i$.*

Proof. We have N independent variables $X_i,\ i = 1, \ldots, N$. Each variable can be in two states (1 or -1) with probability p and $1-p$ respectively. We know the set of admissible values of M. Therefore we have

$$MN \in \{-N, -N+2, -N+4, \ldots, N\},$$

$$NM + N \in \{0, 2, 4, \ldots, 2N\},$$

Now, let us introduce the variable

$$M' = \frac{NM+N}{2} \in \{0, 1, 2, \ldots, N\}.$$

Then M' has Binomial distribution: $M' \in Bin(N,p)$ which is to say

$$P(M' = m) = \binom{N}{m} p^m (1-p)^{N-m},$$

for all $m = 0, 1, \ldots, N$. □

Theorem 3. *The distribution of $\sum_{i<j} X_i X_j$ when $\{X_i\}$ are independent is*

$$P\left(\sum_{i<j} X_i X_j = h\right) = \begin{cases} \binom{N}{\frac{N}{2}} \frac{1}{2^N}, & \text{if } h = -N/2, \\ \binom{N}{\frac{\sqrt{2h+N}+N}{2}} \frac{1}{2^{N-1}}, & \text{if } h \neq -N/2. \end{cases} \tag{3.7}$$

Proof. If $p = \frac{1}{2}$ then
$$P(M' = m') = \binom{N}{m'} \frac{1}{2}^N.$$
So M' is a sum of Bernoulli variables: $M' = \sum_{i=1}^{N} Y_i$, where $Y_i \in \{0,1\}$. Therefore by Lemma (1)
$$\begin{aligned}
\binom{N}{m'} p^{m'}(1-p)^{N-m'} = P(M' = m') &= P\left(\frac{NM+N}{2} = m'\right) \\
&= P\left(M + 1 = \frac{2m'}{N}\right) \\
&= P\left(M = \frac{2m'}{N} - 1\right).
\end{aligned}$$

Let us rename $m = \frac{2m'}{N} - 1$, then $m' = \frac{N(m+1)}{2}$ and
$$P(M = m) = \binom{N}{\frac{N(m+1)}{2}} p^{\frac{N(m+1)}{2}} (1-p)^{\frac{N(1-m)}{2}} = \{\text{if } p = 1/2\} = \binom{N}{\frac{N(m+1)}{2}} \frac{1}{2^N},$$
where $m \in \mathcal{M}$.

We are interested now in finding the value of
$$\sum_{i=1}^{N-1} \sum_{j=i+1}^{N} X_i X_j = \frac{N^2}{2} M^2 - \frac{N}{2}.$$
Let us call this scale Hamiltonian. Consider distribution of M^2
$$P(M^2 = m^2) = P(|M| = m).$$
If $m \neq 0$ then
$$P(M^2 = m^2) = P(M = m) + P(M = -m) = 2P(M = m).$$
If $m = 0$ then
$$P(M^2 = m^2) = P(M = 0) = \binom{N}{\frac{N}{2}} \frac{1}{2^N}.$$
If h is in the set of the admissible values of $\sum_{i=1}^{N-1} \sum_{j=i+1}^{N} X_i X_j$, then
$$\begin{aligned}
P\left(\sum_{i=1}^{N-1} \sum_{j=i+1}^{N} X_i X_j = h\right) &= P\left(\frac{N^2}{2} M^2 - \frac{N}{2} = h\right) \\
&= P\left(\frac{N^2}{2} M^2 = \frac{N}{2} + h\right) \\
&= P\left(M^2 = \frac{2h+N}{N^2}\right).
\end{aligned}$$

$P\left(\sum_{i<j} X_iX_j = h\right) = P(M=0) = \left(\begin{array}{c} N \\ \frac{N}{2} \end{array}\right) \frac{1}{2^N}$, if $h = -N/2$,
$P\left(\sum_{i<j} X_iX_j = h\right) = 2P\left(M = \frac{\sqrt{2h+N}}{N}\right) = \left(\begin{array}{c} N \\ \frac{\sqrt{2h+N}+N}{2} \end{array}\right) \frac{1}{2^{N-1}}$,
if $h \neq -N/2$. Thus we obtain the distribution of $\sum_{i=1}^{N-1}\sum_{j=i+1}^{N} X_iX_j$ as stated. □

3.1.3 Asymptotics

Theorem 4. *For large N*

$$P\left(\sum_{i<j} X_iX_j = h\right) \approx \frac{1}{\sqrt{\pi\,(2h+N)}} e^{-\frac{2h+N}{4}}.$$

Proof. We obtained in Lemma (1) that if $\{X_i\}$, $i = 1, \ldots, N$ are independent the variable $\frac{NM+N}{2} \in Bin(N,p)$, where $p = P(X_i = 1)$. According to the Central Limit theorem the binomial distribution with parameters N and p is approximately normal for large N with parameters $\mu = Np$ and $\sigma^2 = Np(1-p)$. Thus $\frac{NM+N}{2} \in \mathcal{N}(Np, Np(1-p))$. We have

$$\begin{aligned} P(M=m) &= P\left(\frac{N(M+1)}{2} = \frac{N(m+1)}{2}\right) \\ &= \binom{N}{\frac{N(1+m)}{2}} p^{\frac{N(1+m)}{2}} (1-p)^{\frac{N(1-m)}{2}} \\ &\approx \frac{1}{\sqrt{Np(1-p)2\pi}} e^{-\frac{\left(\frac{N(m+1)}{2} - Np\right)^2}{2Np(1-p)}} \\ &= \frac{1}{\sqrt{Np(1-p)2\pi}} e^{-\frac{N\left(\frac{(m+1)}{2} - p\right)^2}{2p(1-p)}}. \end{aligned}$$

If $p = \frac{1}{2}$ then we obtain $P(M=m) = \sqrt{\frac{2}{\pi N}} e^{-\frac{Nm}{2}}$.

We know the property of the normal distribution: If $X \in \mathcal{N}(\mu, \sigma^2)$ and a and b are real numbers, then $aX + b \in \mathcal{N}(a\mu + b, (a\sigma)^2)$. Thus, if we put $a = \frac{2}{N}$, $b = -1$ than $M \in \mathcal{N}\left(2p-1, \frac{4p(1-p)}{N}\right)$ and $\frac{NM}{\sqrt{2}} \in \mathcal{N}\left(\frac{(2p-1)N}{\sqrt{2}}, 2p(1-p)N\right)$. Therefore $\frac{N^2M^2}{2} \in \mathcal{X}_1^2$ and

$$P\left(\sum_{i<j} X_iX_j = h\right) = P\left(\frac{N^2M^2}{2} = h + \frac{N}{2}\right)$$

$$\approx \frac{1}{(2h+N)^{\frac{1}{2}}\,\Gamma\left(\frac{1}{2}\right)} e^{-\frac{2h+N}{4}}.$$

□

3.1.4 Hamiltonian distribution with dependent traders

Let us now consider the case when the decisions of the traders are not independent.

Theorem 5. *The distribution of $\sum_{i<j} X_i X_j$ when $\{X_i\}$ are not independent is*

$$P\left(\sum_{i<j} X_i X_j = h\right) = \begin{cases} \frac{1}{Z} e^{-\frac{\beta}{2}} \binom{N}{\frac{N}{2}}, & \text{if } h = -N/2, \\ \frac{2}{Z} e^{\frac{h\beta}{N}} \binom{N}{\frac{N+\sqrt{N+2h}}{2}}, & \text{if } h \neq -N/2. \end{cases} \tag{3.8}$$

Proof. We have the magnetization

$$M = \frac{1}{N} \sum_{i=1}^{N} X_i.$$

The number of configurations leading to the magnitization m is

$$\mathcal{N}(m) = \binom{N}{N\frac{1+m}{2}} \approx \exp\left(-N\left[\frac{1+m}{2}\ln\frac{1+m}{2} + \frac{1-m}{2}\ln\frac{1-m}{2}\right]\right).$$

Let us enumerate these configurations $x^1, x^2, \ldots, x^{\mathcal{N}(m)}$. We can find the distribution of the magnetization through the distributions of the configurations of the system.

$$\begin{aligned} p(m) &= P(X = x^1) + P(X = x^2) +, \ldots, + P(X = x^{\mathcal{N}(m)}) \\ &= \frac{1}{Z} e^{-\beta\left(-\frac{N}{2}m^2 + \frac{1}{2}\right)} \binom{N}{N\frac{1+m}{2}}. \end{aligned}$$

We know relation between the magnetization and the Hamiltonian

$$\sum_{i=1}^{N-1} \sum_{j=1}^{N} X_i X_j = \frac{N^2}{2} M^2 - \frac{N}{2}.$$

Therefore we can obtain the distribution of the Hamiltonian

$$\begin{aligned} P\left(\sum_{i=1}^{N-1}\sum_{j=1}^{N} X_i X_j = h\right) &= P\left(\frac{N^2}{2}M^2 - \frac{N}{2} = h\right) \\ &= P\left(M^2 = \frac{N+2h}{N^2}\right). \end{aligned}$$

If $h \neq -\frac{N}{2}$ then

$$\begin{aligned} P\left(\sum_{i<j} X_i X_j = h\right) &= 2P\left(M = \frac{\sqrt{N+2h}}{N}\right) \\ &= \frac{2}{Z} e^{\frac{\beta}{2}\left(\frac{N+2h}{N}-1\right)} \binom{N}{\frac{N+\sqrt{N+2h}}{2}} \\ &= \frac{2}{Z} e^{\frac{h\beta}{N}} \binom{N}{\frac{N+\sqrt{N+2h}}{2}}, \end{aligned}$$

if $h = -\frac{N}{2}$ then

$$\begin{aligned} P\left(\sum_{i<j} X_i X_j = h\right) &= P(M=0) \\ &= \frac{1}{Z} e^{-\frac{\beta}{2}} \binom{N}{\frac{N}{2}}. \end{aligned}$$

□

3.1.5 Property of the Hamiltonian and the magnetization

Theorem 6. *The sequence of the Hamiltonians is a Markov chain.*

Proof. Let we denote $\mathcal{X}_t = \{x_t : H(x_1) = h_1, \ldots, H(x_t) = h_t\}$. Then we have

$$\begin{aligned} p_x\left(h_1, \ldots, h_t \left| H_0 = h_0\right.\right) &= \sum_{x \in \mathcal{X}_t} p_x(X_1 = x_1 \left| H_0 = h_0\right.) p_x(x_2 \left| x_1\right.) \cdots p_x(x_t \left| x_{t-1}\right.) \\ &= \sum_{x \in \mathcal{X}_t} Z_{h_0}^{-1} e^{\frac{\beta}{N} H(x_1) h_0} \cdots Z_{h_{t-1}}^{-1} e^{\frac{\beta}{N} H(x_t) h_{t-1}} \\ &= \frac{M_{h_1}}{Z_{h_0}} e^{\frac{\beta}{N} h_1 h_0} \cdots \frac{M_{h_t}}{Z_{h_{t-1}}} e^{\frac{\beta}{N} h_t h_{t-1}}, \end{aligned}$$

where $M_h = \#\{x : H(x) = h\}$.

Then

$$\begin{aligned} \frac{p_x\left(h_t \left| h_0\right.\right)}{p_x\left(h_{t-1} \left| h_0\right.\right)} &= \frac{M_{h_t}}{Z_{h_{t-1}}} e^{\frac{\beta}{N} h_t h_{t-1}} \\ &= \sum_{x : H(x) = h_t} Z_{h_{t-1}}^{-1} e^{\frac{\beta}{N} H(x) h_{t-1}} \\ &= p_x\left(h_t \left| h_{t-1}\right.\right), \end{aligned}$$

or $p_x\left(h_1, \ldots, h_t\right) = p_x\left(h_t \left| h_{t-1}\right.\right) p_x\left(h_1, \ldots, h_{t-1}\right)$. □

Theorem 7. *The sequence of the Magnetizations is a Markov chain.*

Proof. As in the proof of the previous theorem let us denote

$$\mathcal{Y}_t = \{x_t : M(x_1) = m_1, \ldots, M(x_t) = m_t\}.$$

Then we have

$$\begin{aligned} p_x(m_1, \ldots, m_t \,|M_0 = m_0) &= \sum_{x \in \mathcal{Y}_t} p_x(X_1 = x_1 \,|M_0 = m_0) p_x(x_2 \,|x_1) \cdots p_x(x_t \,|x_{t-1}) \\ &= \sum_{x \in \mathcal{Y}_t} Z_{m_0}^{-1} e^{\frac{\beta}{N} \frac{m_1^2 N - 1}{2} \frac{m_0^2 N - 1}{2}} \cdots Z_{m_{t-1}}^{-1} e^{\frac{\beta}{N} \frac{m_t^2 N - 1}{2} \frac{m_{t-1}^2 N - 1}{2}} \\ &= \frac{\mathcal{N}(m_1)}{Z_{m_0}} e^{\frac{\beta}{N} \frac{m_1^2 N - 1}{2} \frac{m_0^2 N - 1}{2}} \cdots \frac{\mathcal{N}(m_t)}{Z_{m_{t-1}}} e^{\frac{\beta}{N} \frac{m_t^2 N - 1}{2} \frac{m_{t-1}^2 N - 1}{2}}, \end{aligned}$$

still $\mathcal{N}(m) = \#\{x : M(x) = m\}$.

Then

$$\begin{aligned} \frac{p_x(m_t \,|m_0)}{p_x(m_{t-1} \,|m_0)} &=^{def} p_x(m_t \,|m_{t-1}, \ldots, m_0) \\ &= \frac{\mathcal{N}(m_t)}{Z_{m_{t-1}}} e^{\frac{\beta}{N} \frac{m_{t-1}^2 N - 1}{2} \frac{m_t^2 N - 1}{2}} \\ &= \sum_{x : M(x) = m_t} Z_{m_{t-1}}^{-1} e^{\frac{\beta}{N} \frac{m(x)^2 N - 1}{2} \frac{m_{t-1}^2 N - 1}{2}} \\ &= p_x(m_t \,|m_{t-1}), \end{aligned}$$

or $p_x(m_t \,|m_{t-1}, \ldots, m_0) = p_x(m_t \,|m_{t-1})$. □

3.1.6 Magnetization distribution with time dependance

We have the process $\{X_{i,t} : \ i = 1, \ldots, N, \ t \in \mathbb{Z}\}$. Let us calculate the conditional probability of the magnetization $P(M_t = m_t \,|M_{t-1} = m_{t-1})$, the probability that the magnetization now is m_t conditional on that it was m_{t-1} at the previous moment.

Theorem 8. $P(M_t = m_t \,|M_{t-1} = m_{t-1}) = \binom{N}{N\frac{1+m_t}{2}} Z_{m_{t-1}}^{-1} e^{\frac{\beta}{N} \frac{1 - N m_t^2}{2} \frac{1 - N m_{t-1}^2}{2}}$.

Proof. The number of configurations leading to the magnetization m_t is $\mathcal{N}(m_t) = \binom{N}{N\frac{1+m_t}{2}}$. Let us enumerate these configurations $x_t^1, x_t^2, \ldots, x_t^{\mathcal{N}(m_t)}$. Correspondly for the magnetization m_{t-1} we have $\mathcal{N}(m_{t-1}) = \binom{N}{N\frac{1+m_{t-1}}{2}}$ configurations: $x_{t-1}^1, x_{t-1}^2, \ldots, x_{t-1}^{\mathcal{N}(m_{t-1})}$ and so on.

Thus,

$$P(M_t = m_t \,|M_{t-1} = m_{t-1}\,) =$$

$$\begin{aligned}
&= \frac{P(M_t = m_t, \ldots, M_1 = m_1 \,|M_0 = m_0\,)}{P(M_{t-1} = m_{t-1}, \ldots, M_1 = m_1 \,|M_0 = m_0\,)} \\
&= \frac{\sum_{l_1=1}^{\mathcal{N}(m_1)} \cdots \sum_{l_t=1}^{\mathcal{N}(m_t)} P\left(X_t = x_t^{l_t} \,\middle|\, X_{t-1} = x_{t-1}^{l_{t-1}}\right) \cdots P(X_1 = x_1^{l_1} \,|M_0 = m_0)}{\sum_{l_1=1}^{\mathcal{N}(m_1)} \cdots \sum_{l_{t-1}=1}^{\mathcal{N}(m_{t-1})} P\left(X_{t-1} = x_{t-1}^{l_{t-1}} \,\middle|\, X_{t-2} = x_{t-2}^{l_{t-2}}\right) \cdots P(X_1 = x_1^{l_1} \,|M_0 = m_0)} \\
&= \frac{\mathcal{N}(m_1)\cdots\mathcal{N}(m_t)P(X_t = x_t \,|X_{t-1} = x_{t-1}) \cdots P(X_1 = x_1 \,|M_0 = m_0)}{\mathcal{N}(m_1)\cdots\mathcal{N}(m_{t-1})P(X_{t-1} = x_{t-1} \,|X_{t-2} = x_{t-2}) \cdots P(X_1 = x_1 \,|M_0 = m_0)} \\
&= \mathcal{N}(m_t)P(X_t = x_t \,|X_{t-1} = x_{t-1}) \\
&= \binom{N}{N\frac{1+m_t}{2}} Z_{m_{t-1}}^{-1} e^{\frac{\beta}{N}\frac{1-Nm_t^2}{2}\frac{1-Nm_{t-1}^2}{2}}.
\end{aligned}$$

□

3.1.7 Hamiltonian distribution with time dependance

Let us consider conditional distribution of the Hamiltonian in time.

Theorem 9. *The conditional distribution of* $\sum_{i=1}^{N-1}\sum_{j=i+1}^{N} X_i X_j$ *is*

$$P\left(\sum_{i<j} X_{i,t}X_{j,t} = h_t \,\middle|\, \sum_{i<j} X_{i,t-1}X_{j,t-1} = h_{t-1}\right) =$$

$$= \begin{cases} 2\binom{N}{\frac{N+\sqrt{N+2h_t}}{2}} Z_{h_{t-1}}^{-1} e^{\frac{\beta}{N^3}h_t h_{t-1}}, & \textit{if } h_t \neq -\frac{N}{2}, \\ \binom{N}{\frac{N}{2}} Z_{h_{t-1}}^{-1} e^{-\frac{h_{t-1}\beta}{2N^2}}, & \textit{if } h_t = -\frac{N}{2}. \end{cases}$$

Proof. Let us calculate the number of configurations leading to the Hamiltonian magnetization h. Remember that $h = \frac{1-Nm^2}{2}$ we have

$$\begin{aligned}
M_h &= \begin{cases} 2\mathcal{N}(m), & \text{if } m \neq 0, \\ \mathcal{N}(0), & \text{if } m = 0, \end{cases} \\
&= \begin{cases} 2\binom{N}{N\frac{1+m}{2}}, & \text{if } m \neq 0, \\ \binom{N}{\frac{N}{2}}, & \text{if } m = 0, \end{cases}
\end{aligned}$$

$$= \begin{cases} 2\begin{pmatrix} N \\ \frac{N+\sqrt{N(1-2h)}}{2} \end{pmatrix}, & \text{if } h \neq \frac{1}{2}, \\ \begin{pmatrix} N \\ \frac{N}{2} \end{pmatrix}, & \text{if } h = \frac{1}{2}. \end{cases}$$

Let us enumerate the configurations leading to the Hamiltonian $h_t : x_t^1, x_t^2, \dots, x_t^{M_{h_t}}$. Correspondly for the Hamiltonian h_{t-1} we have the configurations: $x_{t-1}^1, x_{t-1}^2, \dots, x_{t-1}^{M_{h_{t-1}}}$ and so on.
Thus,

$$P(H_t = h_t \,|H_{t-1} = h_{t-1}\,) =$$

$$\begin{aligned}
&= \frac{P(H_t = h_t, \dots, H_1 = h_1 \,|H_0 = h_0\,)}{P(H_{t-1} = h_{t-1}, \dots, H_1 = h_1 \,|H_0 = h_0\,)} \\
&= \frac{\sum_{l_1=1}^{M_{h_1}} \cdots \sum_{l_t=1}^{M_{h_t}} P\left(X_t = x_t^{l_t} \,\middle|\, X_{t-1} = x_{t-1}^{l_{t-1}}\right) \cdots P(X_1 = x_1^{l_1} \,|H_0 = h_0)}{\sum_{l_1=1}^{M_{h_1}} \cdots \sum_{l_{t-1}=1}^{M_{h_{t-1}}} P\left(X_{t-1} = x_{t-1}^{l_{t-1}} \,\middle|\, X_{t-2} = x_{t-2}^{l_{t-2}}\right) \cdots P(X_1 = x_1^{l_1} \,|H_0 = h_0)} \\
&= \frac{M_{h_1} \cdots M_{h_t} P(X_t = x_t \,|X_{t-1} = x_{t-1}) \cdots P(X_1 = x_1 \,|H_0 = h_0)}{M_{h_1} \cdots M_{h_{t-1}} P(X_{t-1} = x_{t-1} \,|X_{t-2} = x_{t-2}) \cdots P(X_1 = x_1 \,|H_0 = h_0)} \\
&= M_{h_t} P(X_t = x_t \,|X_{t-1} = x_{t-1}) \\
&= \begin{cases} 2\begin{pmatrix} N \\ N\frac{1+\sqrt{\frac{1-2h_t}{N}}}{2} \end{pmatrix} Z_{h_{t-1}}^{-1} e^{\frac{\beta}{N} h_t h_{t-1}}, & \text{if } h_t \neq \frac{1}{2}, \\ \begin{pmatrix} N \\ \frac{N}{2} \end{pmatrix} Z_{h_{t-1}}^{-1} e^{\frac{h_{t-1}\beta}{2N}}, & \text{if } h_t = \frac{1}{2}, \end{cases}
\end{aligned}$$

where $Z_{h_{t-1}} = \sum_{x_t} e^{\frac{\beta}{N} h_{t-1} H(x_t)}$,

$$P\left(\sum_{i<j} X_{i,t} X_{j,t} = h_t \,\middle|\, \sum_{i<j} X_{i,t-1} X_{j,t-1} = h_{t-1}\right) =$$

$$\begin{aligned}
&= P\left(H_t = -\frac{h_t}{N} \,\middle|\, H_{t-1} = -\frac{h_{t-1}}{N}\right) \\
&= \begin{cases} 2\begin{pmatrix} N \\ \frac{N+\sqrt{N+2h_t}}{2} \end{pmatrix} Z_{h_{t-1}}^{-1} e^{\frac{\beta}{N^3} h_t h_{t-1}}, & \text{if } h_t \neq -\frac{N}{2}, \\ \begin{pmatrix} N \\ \frac{N}{2} \end{pmatrix} Z_{h_{t-1}}^{-1} e^{-\frac{h_{t-1}\beta}{2N^2}}, & \text{if } h_t = -\frac{N}{2}. \end{cases}
\end{aligned}$$

where $Z_{h_{t-1}} = \sum_{x_t} e^{\frac{\beta h_{t-1}}{N^3} H(x_t)}$ now.

$\square$

3.1.8 Conditional expectation and variance of the Hamiltonian

Theorem 10. *The conditional expectation of the Hamiltonian is*

$$E\left(H(X_t)\left|H(X_{t-1})\right.\right) = \frac{N}{H_{t-1}}\frac{d\ln Z(\beta)}{d\beta}.$$

Proof. We have $Z = \sum_{X_t} e^{\frac{\beta}{N}H(X_t)H(X_{t-1})}$. Then

$$\begin{aligned}
\frac{d\ln Z(\beta)}{d\beta} &= \frac{1}{Z(\beta)}\sum_{X_t} e^{H_tH_{t-1}\beta/N}H(X_t)H(X_{t-1})/N \\
&= \frac{H(X_{t-1})}{N}\frac{1}{Z(\beta)}\sum_{X_t} e^{H_tH_{t-1}\beta/N}H(X_t).
\end{aligned}$$

From the other side we know the mathimatical conditional expectation of the Hamiltonian

$$E\left(H(X_t)\left|H(X_{t-1})\right.\right) = \sum_{X_t} H(X_t)p\left(X_t\left|X_{t-1}\right.\right) = \frac{1}{Z}\sum_{X_t} e^{H_tH_{t-1}\beta/N}H(X_t).$$

□

Theorem 11.

$$Var\left(H(X_t)\left|H(X_{t-1})\right.\right) = \left(\frac{N}{H_{t-1}}\right)^2\frac{d^2\ln Z(\beta)}{d\beta^2}.$$

Proof. From the previous theorem we have

$$\begin{aligned}
\frac{d^2\ln Z(\beta)}{d\beta^2} &= \frac{d}{d\beta}\left(\frac{H_{t-1}}{N}\frac{\sum_{X_t} H_t e^{H_tH_{t-1}\beta/N}}{Z}\right) \\
&= \frac{H_{t-1}}{N}\left(\frac{\sum_{X_t} H_t^2\frac{H_{t-1}}{N}e^{H_tH_{t-1}\beta/N}}{Z} - \frac{\left(\sum_{X_t} H_t e^{H_tH_{t-1}\beta/N}\right)^2}{Z^2}\right) \\
&= \left(\frac{H_{t-1}}{N}\right)^2\left[E\left(H_t^2\left|H_{t-1}\right.\right) - \left(E\left(H_t\left|H_{t-1}\right.\right)\right)^2\right] \\
&= \left(\frac{H_{t-1}}{N}\right)^2 Var\left(H_t\left|H_{t-1}\right.\right).
\end{aligned}$$

□

Theorem 12.

$$E\left(H_t\left|H_{t-1}\right.\right) \approx \frac{N\left(1-2H_{t-1}\right)}{4H_{t-1}\left(N-\frac{\beta}{2}\left(1-2H_{t-1}\right)\right)}.$$

Proof. Remember the value of the partition function

$$Z(\beta) \approx \exp(N\ln 2)\sqrt{\frac{2\pi}{N\left(1-\frac{\beta}{2}\frac{1-2H_{t-1}}{N}\right)}},$$

calculate

$$\begin{aligned}
\frac{d\ln Z}{d\beta} &= \frac{d}{d\beta}\ln\left(\sqrt{\frac{2\pi}{N\left(1-\frac{\beta}{2}\frac{1-2H_{t-1}}{N}\right)}}\right) \\
&= \frac{1}{\sqrt{\frac{2\pi}{N\left(1-\frac{\beta}{2}\frac{1-2H_{t-1}}{N}\right)}}}\frac{1}{2}\frac{1}{\sqrt{\frac{2\pi}{N\left(1-\frac{\beta}{2}\frac{1-2H_{t-1}}{N}\right)}}}\frac{-2\pi N\left(-\frac{1}{2}\frac{1-2H_{t-1}}{N}\right)}{\left(N\left(1-\frac{\beta}{2}\frac{1-2H_{t-1}}{N}\right)\right)^2} \\
&= \frac{\pi N\left(1-\frac{\beta}{2}\frac{1-2H_{t-1}}{N}\right)\left(1-2H_{t-1}\right)}{4\pi\left[N\left(1-\frac{\beta}{2}\frac{1-2H_{t-1}}{N}\right)\right]^2} \\
&= \frac{1-2H_{t-1}}{4\left(N-\frac{\beta}{2}\left(1-2H_{t-1}\right)\right)}.
\end{aligned}$$

And

$$E\left(H_t\left|H_{t-1}\right.\right) = \frac{N}{H_{t-1}}\frac{d\ln Z}{d\beta} = \frac{N\left(1-2H_{t-1}\right)}{4H_{t-1}\left(N-\frac{\beta}{2}\left(1-2H_{t-1}\right)\right)}.$$

□

Theorem 13.

$$Var\left(H_t\left|H_{t-1}\right.\right) = \frac{N^2\left(1-2H_{t-1}\right)^2}{8H_{t-1}^2\left(N-\frac{\beta}{2}\left(1-2H_{t-1}\right)\right)^2}.$$

Proof. To use the theorem above we calculate

$$\begin{aligned}
\frac{d^2\ln Z}{d\beta^2} &= \frac{d}{d\beta}\left(\frac{1-2H_{t-1}}{4\left(N-\frac{\beta}{2}\left(1-2H_{t-1}\right)\right)}\right) \\
&= \frac{-\left(1-2H_{t-1}\right)4\left[-\frac{1}{2}\left(1-2H_{t-1}\right)\right]}{\left[4\left(N-\frac{\beta}{2}\left(1-2H_{t-1}\right)\right)\right]^2} \\
&= \frac{\left(1-2H_{t-1}\right)^2}{8\left(N-\frac{\beta}{2}\left(1-2H_{t-1}\right)\right)^2}.
\end{aligned}$$

Thus

$$Var\left(H_t\left|H_{t-1}\right.\right)=\frac{N^2\left(1-2H_{t-1}\right)^2}{8H_{t-1}^2\left(N-\frac{\beta}{2}\left(1-2H_{t-1}\right)\right)^2}.$$

□

Therefore we can obtain the limit of the conditional expectation and vatiance $\lim_{N\to\infty} E\left(H_t\left|H_{t-1}\right.\right) = \lim_{N\to\infty} \frac{N(1-2H_{t-1})}{4H_{t-1}N\left(1-\frac{\beta}{2N}(1-2H_{t-1})\right)} = \frac{1-2H_{t-1}}{4H_{t-1}}$ $\lim_{N\to\infty} Var\left(H_t\left|H_{t-1}\right.\right) = \frac{(1-2H_{t-1})^2}{8H_{t-1}^2}$. The Hamiltonian has the normal distribution for large N, so

$$p\left(H_t=h_t\left|H_{t-1}=h_{t-1}\right.\right)=\frac{1}{\sqrt{2\pi}H_{t-1}2\sqrt{2}}\exp\left(\frac{-h_t-\frac{1-2h_{t-1}}{4h_{t-1}}}{16h_{t-1}^2}\right).$$

3.1.9 Stationary distribution

Statement. The process $\{X_t\}$ is a time-homogeneous Markov chain because

$$P(X_{t+1}=x\left|X_t=y\right.)=P(X_t=x\left|X_{t-1}=y\right.).$$

If Π is stationary distribution and $\{X_t\}$ is a time-homogeneous Markov Chain then

$$\Pi(\mathbb{X})=\Pi(\mathbb{Y})P,$$

where P is a matrix of the transition probabilities.

Theorem 14. *The process $\{X_t\}$ is time-reversible and it has stationary distribution $\Pi(x)=\frac{Z(x)}{\sum_x Z(x)}$.*

Proof. We have that

$$\Pi(x)=\sum_y \Pi(y)p(x\left|y\right.).$$

Let us try to find Π such that

$$\Pi(y)p(y\left|x\right.)=\Pi(x)(x)p(x\left|y\right.). \tag{3.9}$$

The existance of such Π will mean that the Markov chain is reversible. We can rewrite Equation (3.9)

$$\frac{\Pi(y)e^{\beta H(y)H(x)}}{\sum_y e^{\beta H(y)H(x)}}=\frac{\Pi(x)e^{\beta H(x)H(y)}}{\sum_x e^{\beta H(x)H(y)}},$$

or

$$\frac{\Pi(y)}{Z(y)} = \frac{\Pi(x)}{Z(x)}.$$

This implies $\frac{\Pi(x)}{Z(x)} = c$, where c is some constant with respect to x. $\Pi(x) = Z(x)c$. Further $\sum_x \Pi(x) = c \sum_x Z(x) = 1$ and $c = \frac{1}{\sum_x Z(x)}$, so we finally obtain that $\Pi(x) = \frac{Z(x)}{\sum_x Z(x)}$. □

3.2 Exact calculations

We obtained the exact distribution of the $\sum_{i<j} X_i X_j$. Now let us verify this distribution for 10 traders. We wrote a program in the R language of programming which calculates the probability of all possible configurations of the system containing N traders, by determining the value of the Hamiltonian for each configuration and build the matrix `Vec` of dimension $3 \times$ ("the largest possible value of H" $-$ "the smallest possible value of H") where the first line contains all possible values of H, the second line: the number of the configurations that leads to the corresponding value in the first line, the third: probability of corresponding value from the first line. In Figure 3.3 we can see the distribution of H for $N = 10$.

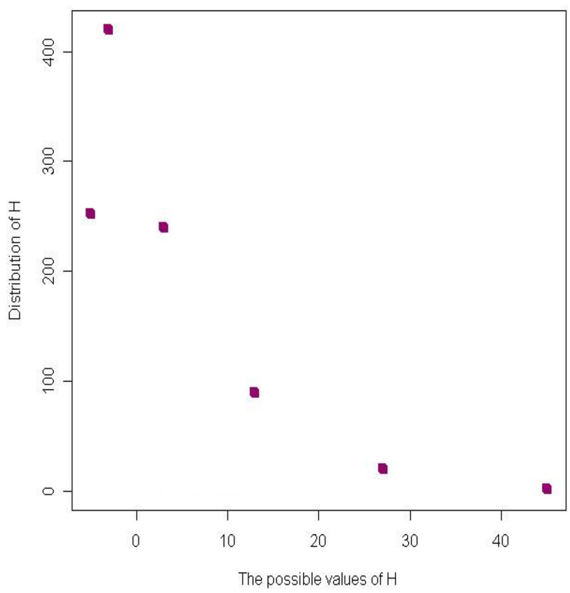

Figure 3.3: *The exact distribution of the Energy function for ten traders.*

Table 3.1: *The exact distribution of the Hamiltonian for 10 traders*

Place	-5	-3	3	13	27	45
Number of occurance	252	420	240	90	20	2
Probability	0.246	0.41	0.234	0.088	0.0195	0.00195

In the Table 3.1 bellow we can see these result.
Let us verify the exact distribution using formulas above and the values in the table for $N = 10$. We obtain

$$P\left(\sum_{i<j} X_i X_j = -5\right) = P(M=0) = \binom{10}{5}\frac{1}{2^{10}} =$$

$$= \frac{10!}{5!5!2^{10}} = \frac{10\cdot 9\cdot 8\cdot 7\cdot 6}{5\cdot 4\cdot 3\cdot 2\cdot 2^{10}} = \frac{63}{256} = 0.246.$$

The value '-4' is not admissible.

$$P\left(\sum_{i<j} X_i X_j = -3\right) = 2\cdot P\left(M = \frac{\sqrt{-3\cdot 2+10}}{10}\right) = 2\cdot P\left(M = \frac{1}{5}\right) =$$

$$= \left(\begin{array}{c} 10 \\ \frac{\sqrt{-3\cdot 2+10}+10}{2} \end{array}\right)\frac{1}{2^9} = \binom{10}{6}\frac{1}{2^9} = \frac{10!}{6!4!2^9} = \frac{10\cdot 9\cdot 8\cdot 7}{4\cdot 3\cdot 2\cdot 2^9} = \frac{105}{256} = 0.41.$$

For other values computations are similar. So, the exact distribution is confirmed.

3.3 The hypothesis testing

Test of independence

If the Hamiltonian has distribution like in the previous section then trader's actions are almost independent and β is small. Otherwise we can talk about magnetization, which could be the reason of bubbles and crashes on the market. It is important to see if the value of β deviates from zero to such an extent that dangerous development is indicated. Then β can become close to the critical value and bubble or crash on the market can appear. To calculate the difference between the exact didtribution and real data we will use the function $\sum_{i=1}^{K}\frac{(O_i-E_i)^2}{E_i} \sim \mathcal{X}_K^2$, where O_i is the observed number of occurrance in class i, E_i is the expected number of occurrance in class i and K is the number of classes.

We want to test the hypothesis

$$\begin{cases} H_0: \ \beta = 0, \\ H_1: \ \beta \neq 0. \end{cases}$$

For each interval of time we calculate the value of the Hamiltonian and then the probability of each value.

Example
Let us consider the data from the Swedish steel market for the October 22, 2008. We have the traders and buyers with certain moment of trading. To analyze these data we will collect information about trading by dividing the time on the parts each about 10 minutes. We have 10 the most active traders (AVA, CSB, DBL, ENS, EVL, MSI, NDS, NON, SHB, SWB) and 20 intervals of activity. Let us consider the Hamiltonian for these traders. The result is in the following Table 3.2.

Table 3.2: *The results of the observations*

Place	-5	-3	3	13	27	45
Number of occurance	2	13	4	1	0	0
Expected number of occurance	4.92	8.2	4.688	1.76	0.39	0.04

Thus $N = 10$ and $K = 6$, so the value of the test ststistic in this case is

$$\frac{(2-4.92)^2}{4.92} + \frac{(13-8.2)^2}{8.2} + \frac{(4-4.688)^2}{4.688} + \frac{(1-1.76)^2}{1.76} + 0.39 + 0.04 = 5.39.$$

The value $5.39 < 12.5916 = \mathcal{X}_6^2(0.05)$, where N_H is the number of occurance in the test, N_E is expected number of occurance.

We can see that distribution of the Hamiltonian which was obtained in the hypothesis test and distribution of the Hamiltonian in the case of independence of the traders are almost independent.

Chapter 4

Conclusions

In this thesis we investigated a new space-time model for interacting agents in the financial market. First we reviewed the history of the Ising model and some other Ising-type models. Then the Ising model, Curie-Weiss model and some modifications of these models were formally presented. Also we considered one way of finding the critical temperature of the market. A new space-time model was developed and necessary and sufficient conditions for its stationarity were found. The non-linear sensitivity of market global properties in terms of temperature parameter changes was investigated. The critical temperature for this model was analytically derived.

The distribution of the Hamiltonian was analyzed using its dependence with the magnetization of the market and the exact distribution was calculated. The conditional expectation and variance of the Hamiltonian were found and the stationary distribution was obtained. Then the exact distribution of the Hamiltonian for 10 traders was calculated, and the expected distribution was confirmed. Hypothesis test for independence between agents was considered for the Swedish steel market and it showed that there is no evident critical situation on the market at the time of this dataset.

The parameter β reflects how strongly traders are influenced by each other in the market. It can signalize risk for a crash or a bubble in the market. Therefore it is very important that its analysis is accurate in such situations. What remains for future work? We could try a lot of real data and compare inferential results relying on this model to observable quantities generally accepted as a measure of health of the situation on the market. Also we can consider a bigger ammount of traders to have a more exact p-value in the hypothesis testing. Then we can estimate the amount of interaction (i.e. β) for our model using, e.g. maximum likelihood estimator. Here exists also the possibility to develop hypothesis testing based on a time dependent model. Also interesting to find out how good this model is to explain volatility clustering.

Notation

N	an amount of the traders.
i	a number of the trader, $i = 1, \dots, N$.
X_i	a decision of the trader i to buy or to sell the stock, $X_i \in \{-1, 1\}$.
p	a probability that the decision of the trader is $+1$ (to buy).
$X = (X_1, X_2, \dots, X_N)$	an investment attitude of the market.
$X_t = (X_{1,t}, X_{2,t}, \dots, X_{N,t})$	an investment attitude of the market at the time t.
$i \sim j$	a neighborhood relation between traders i and j.
$x = \{x_i : i = 1, 2, \dots, N\}$	a confguration of the system.
K	a number of the distinct configurations.
$p(x)$	a probability of the configuration x.
$\aleph = \{-1, 1\}^N$	the set of all configurations of the system.
Z	a partition function.
β	temperature of the market.
$q_i(x)$	a disagreement function of the trader i in the configuration x.
$H(x)$	Hamiltonian of the configuration x.

$M(X) = \sum_{i=1}^{N} X_i$	a magnetization of the system.
$m(x) = \sum_{i=1}^{N} x_i$	a magnetization of the configuration x.
d	a size of the stock.
Λ	a symmetric hypercube.
D	the dimension of the hypercube Λ.
s	an investment environment.
Q	an entropy of the system.
$\Pi(X)$	a stationary distribution of the process $\{X\}$.
$E_P(X)$	an expectation of the random variable X with respect to the measure P.
$Var_P(X)$	a variance of the random variable X with respect to the measure P.
$\mathcal{M}$	the set of the admissible values of the magnetization.
$\delta_{mx} = \begin{cases} 1, & \text{if } m(x) = m \\ 0, & \text{if } m(x) \neq m \end{cases}$	Kronecker's delta.
$\mathcal{N}(m)$	a total number of all configurations leading to the magnetization m.

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Appendix

Code of the program in R

Function *num* calculates the size of the binomial number that corresponds to the number i.

```
num<-function(i)
  {
    len=0
    while(2^len<=i){len=len+1}
    len
  }
```

Function $num{-}bin$ forwards nonnegative integer number i in the binomial number which is presented in the form of the vector with length *len*

```
num-bin <-function(i,len)
  {
    for (j in 1:len)
    {
      ch=i/2
      ch_round=floor(ch)
      x[k-j+1]=i-2*ch_round
      i=ch_round
    }
    x
  }

Nn=10
k=Nn
N=2^Nn
x<-1:k
if((Nn/2-floor(Nn/2))==0){Len_Q=-Nn/2} else {Len_Q=(-Nn+1)/2}
Vec<-matrix(0,2,-Len_Q+(Nn*(Nn-1)/2)+1)
```

```
for(ti in 1:floor(-Len_Q+(Nn*(Nn-1)/2)+1))Vec[1,ti]=Len_Q+ti-1
for(i in 1:N)
  {
    x<-2*num_bin(i-1,k)-1
    sm=0
    for(j in 1:(k-1))
      {
      for(z in (j+1):k)
        {sm=sm+x[j]*x[z]}
      }
  Vec[2,sm+1-Len_Q]=Vec[2,sm+1-Len_Q]+1
  }
```